

1. Connect a USB webcam to your computer, install any required drivers, and verify that the webcam works by opening the default camera app and capturing a photo.

- **Step 1: Connect the Webcam** — Plug the USB cable directly into an available USB port on your desktop or laptop.
- **Step 2: Install Drivers** — Windows 10/11 and macOS automatically install standard UVC drivers within a few seconds. If your device came with dedicated software (like Logitech G HUB), install that next.
- **Step 3: Open the Camera App**
 - **Windows:** Press Win + S, type **Camera**, and press **Enter**.
 - **macOS:** Open **Finder** → **Applications** → **Photo Booth**.
- **Step 4: Capture a Photo** — Switch the active camera input to your external USB webcam if needed, align your shot, and click the camera icon to snap a photo.

2.Set up a local printer (USB or network), install the correct driver, and print a test page to confirm successful installation.

Step 1: Physical Setup & Connection

For a USB Printer:

1. Connect the USB cable from the printer directly into an available USB port on your computer.
2. Turn on the printer's power switch.

For a Network Printer (Wi-Fi or Ethernet):

1. Ethernet: Connect an Ethernet cable from your printer directly to your Wi-Fi router.
2. Wi-Fi: On the printer's control panel, go to Network Settings → Setup Wizard (or Wi-Fi Setup), select your local Wi-Fi network, and enter the password.
3. Ensure your computer is connected to the same Wi-Fi/local network as the printer.

Step 2: Driver Installation

Modern operating systems usually detect and install standard drivers automatically:

Windows:

1. Open Settings (Win + I).

2. Go to Bluetooth & devices → Printers & scanners (or Devices → Printers & scanners on Windows 10).
3. Click Add device (or Add a printer or scanner).
4. Windows will scan your local network/USB ports and list detected printers. Select your printer and click Add device.

macOS:

1. Open System Settings (or System Preferences) → Printers & Scanners.
2. Click Add Printer, Scanner, or Fax... (or the + button).
3. Select your printer from the list (it will appear under Default or Network tabs via AirPrint/Bonjour). Click Add.

Step 3: Print a Test Page

Printing a test page verifies proper communication between your operating system and the printer hardware.

On Windows:

1. Return to Settings → Printers & scanners.
2. Click on your newly installed printer.
3. Click Printer properties (or Manage → Printer properties).
4. Under the General tab, click the Print Test Page button at the bottom right.

On macOS:

1. Open System Settings → Printers & Scanners.
2. Click on your installed printer.
3. Click Options & Supplies... → select the Utility tab.
4. Click Print Test Page (or Open Printer Utility to run a nozzle/test check).

3. Simulate a connectivity issue by unplugging the printer's USB cable, then use Windows Device Manager to identify and resolve the 'printer not detected' problem.

Hint: Check for error icons and use the 'Scan for hardware changes' option after reconnecting.

1. Simulate the Disconnection

1. **Unplug the Cable:** Disconnect the printer's USB cable from your computer.
2. **Observe the Effect:** Open **Settings → Printers & scanners**. The printer status will change to "**Offline**" or "**Not connected**".

2. Using Device Manager to Identify the Issue

When a printer is unplugged or experiencing a port connection error, Device Manager reflects this status:

1. **Open Device Manager:**
 - Press Win + X and select **Device Manager** (or press Win + R, type devmgmt.msc, and press **Enter**).
2. **Look for Status Changes or Error Icons:**
 - **Hidden / Grayed Out Devices:** Expand **Print queues** or **Printers**. Unplugged USB devices will often disappear from the active list. (To view them, click **View** in the top

menu bar and select **Show hidden devices**—the disconnected printer icon will appear faded/grayed out).

- **Warning Icons:** If there is a driver or communication glitch caused by the sudden drop, you may see a **yellow exclamation mark (!)** or **red X** next to the printer name or under **Universal Serial Bus controllers** (USB Printing Support).

3. Resolving the Problem (Reconnection & Hardware Rescan)

1. **Reconnect the Cable:** Plug the USB cable back securely into your computer's USB port (try a direct port on the motherboard/chassis rather than an unpowered hub).

2. Trigger a Hardware Rescan:

- In Device Manager, click on your computer's name at the very top of the device list.
- Click **Action** in the top menu bar, then click **Scan for hardware changes** (or click the monitor icon with a magnifying glass on the toolbar).

3. Verify Resolution:

- Device Manager will refresh its tree view.
- The **Print queues** section will automatically expand, showing your printer as active (no longer grayed out and without any warning icons).

- Check **Printers & scanners** in Windows Settings—the status should return to "**Ready**".

4.A user reports that their webcam video is not displaying in Zoom, but it works in the default camera app. Troubleshoot and resolve the issue so the webcam works in both applications.

Hint: Check if another app is using the webcam or if Zoom permissions are enabled.

Step 1: Close the Camera app:

Webcams can only work in one app at a time. Close the Camera app (and any other video apps like Teams or Skype), then open Zoom again.

Step 2: Allow Zoom to use your camera:

- On Windows: Go to Settings > Privacy & security > Camera. Turn ON Camera access and make sure Zoom is allowed to use it.
- On Mac: Go to System Settings > Privacy & Security > Camera, and turn ON the toggle switch next to Zoom.

Step 3: Pick the right camera in Zoom:

1. Open Zoom and click your profile picture at the top right, then click Settings.
2. Click Video on the left menu.
3. In the Camera dropdown list, select your webcam's actual name.